

Machine Learning Techniques for Phishing Attack Identification: A Comprehensive Approach of Evaluating Classifiers on Real - World URLs

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Abstract

Phishing websites are a major and fast-growing cyber threat, often changing their structure, domains, and appearance to bypass traditional blacklist and rule-based security systems. This paper presents a machine-learning-based detection framework designed to identify malicious URLs using a combination of lexical features, domain-based information, and content-level behavioral signals. Using a large and diverse dataset of over 700,000 labeled URLs, the system applies feature engineering, dimensionality reduction, and a comparison of several advanced classifiers, including Random Forest, Gradient Boosting, Support Vector Machines, and Neural Networks. Results show that ensemble methods, especially Random Forest, provide the highest accuracy rate of 96% or 0.96, improved detection of phishing URLs, and strong resistance to adversarial evasion techniques. The study demonstrates that intelligent, data-driven approaches can significantly strengthen modern cybersecurity systems by offering scalable and real-time phishing detection.

Index Terms

Phishing Detection, Machine Learning, URL Intelligence, Lexical Feature Analysis, Domain Metadata, Ensemble Learning, Random Forest, Gradient Boosting, Cyber Threat Intelligence, Adversarial Evasion, Real-Time Web Security, Feature Engineering, Malicious URL Classification, Cybersecurity Analytics.

INTRODUCTION

The rapid expansion of digital ecosystems and the ubiquitous reliance on web-based services have drastically amplified the attack surface for cyber-threat actors. Among these, phishing websites represent one of the most pervasive and sophisticated forms of social engineering, engineered to deceptively replicate legitimate platforms with the intent of harvesting user credentials, financial data, and personally identifiable information. Owing to their dynamic nature, polymorphic URL structures, and the rapid churn rate of malicious domains, phishing attacks continue to bypass conventional blacklist-based filters, heuristic scanners, and rule-driven security gateways [14], [15].

Traditional blacklist-based and signature-based approaches are reactive and cannot cope with newly emerging phishing patterns [11]. In contrast, machine learning can generalize examples to detect new phishing URLs through intrinsic lexical and structural patterns [1], [2]. This work presents the development and evaluation of a machine learning-based phishing detection system using multiple datasets combined for high diversity and coverage of real-world phishing techniques. A wide range of classifiers, including linear, probabilistic, tree-based, and

ensemble models, are applied and compared to find an effective method for large-scale phishing detection [3], [4], [10].

This paper is organized as follows: Section I reviews key advancements in phishing URL detection, covering ML/DL models, ensemble strategies. Section II reviews prior to research on URL-based phishing detection. Section III presents the combined dataset including the Malicious URL Dataset, ISCX-URL2016, and sources from Phish Storm, and explains the preprocessing steps. Section IV describes feature engineering that includes lexical, domain-based, and keyword features, along with SMOTE for class balancing. Section V describes model training and evaluation methodology, covering Lazy Classifier benchmarking, comprehensive classifier analysis, and the Soft Voting Ensemble.

Section VI presents the evaluation of the Random Forest model with its performance metrics. Section VII presents experimental results and classifier performance comparison. Finally, in Section VIII, the paper concludes the study and suggests possible research directions in the future.

I. Related Work

Several studies have explored machine learning and deep learning techniques for phishing URL detection. Aslam et al. proposed a stacked generalization model combining TF-IDF, LSTM, and meta-XGBoost, achieving 96.04% accuracy on benchmark datasets [7]. Manerikar et al. introduced URLTran, a transformer-based model that improved true positive rates at extremely low false positive rates [8]. Comparative analyses have shown that Random Forest classifiers, when tuned appropriately, can reach near-perfect accuracy levels [1], [3].

Parallel and sequential architectures have also been evaluated, with XGBoost and MLPs achieving ~90% accuracy [6]. Deep learning optimization techniques such as EGSO-CNN and VAE-based feature extraction have further pushed detection accuracy beyond 99% [3], [5]. Systematic reviews of phishing detection methods highlight the growing role of ensemble learning and feature-level fusion in improving robustness [14][15].

Our work builds upon these foundations by integrating multiple classifiers through stacking and soft voting, guided by TP/TN metrics and bias-variance analysis, to

create a scalable and high-performing phishing detection model.

II. Literature Survey

Albishri and Dessouky (2024) conducted a comprehensive benchmarking study evaluating the performance of classical machine learning algorithms on phishing URL datasets [1]. Their work places strong emphasis on feature engineering and handling class imbalance, ultimately identifying Random Forest—after extensive hyperparameter optimization—as the most accurate model, achieving 99.93–99.98% accuracy. The study highlights the robustness of ensemble-based classifiers in detecting complex phishing patterns but remains limited to traditional ML, without the integration of deep learning or automated model selection frameworks. This study forms a foundational comparative baseline for modern phishing detection architectures.

Aldaham and colleagues (2024) proposed an enhanced hybrid machine learning model optimized for detecting phishing websites using over 30 lexical, content-based, and host-based features [2]. Their framework integrates multiple ML algorithms and feature selection strategies to improve prediction precision while reducing false positives. The model demonstrates high performance on benchmark datasets, particularly due to systematic feature extraction and ensemble fusion. However, the study focuses primarily on static feature-based detection, without addressing real-time adaptability or large-scale URL stream classification.

Barik et al. (2025) introduced the EGSO-CNN model, which leverages Enhanced Grid Search Optimization alongside Convolutional Neural Networks to improve phishing URL detection accuracy [3]. A notable contribution of this work is the integration of Variational Autoencoders (VAE) for automatic feature extraction, enabling the model to learn hidden patterns within URLs more effectively than handcrafted feature approaches. With an accuracy of 99.44%, the model demonstrates the potential of deep learning-based optimization techniques. However, the study's computational complexity is higher, making real-time implementations challenging.

Paithane (2025) proposed URLGuard, a hybrid ensemble model combining Random Forest and Gradient Boosting for phishing detection [4]. The system focuses on enhancing precision-recall performance through complementary strengths of tree-based learners. URLGuard also emphasizes the integration of lexical and host-level features to improve generalization across unseen URLs. While effective, the model relies heavily on

handcrafted features and lacks the deep representational learning provided by neural architectures. Nevertheless, the study demonstrates the benefit of hybrid ensemble stacking for improved detection reliability.

Zhang et al. (2025) explored predictive phishing detection using graph-based machine learning and anomaly detection to identify malicious campaigns before they become active [5]. By analyzing domain relationships, hosting patterns, and temporal behaviors, the model achieved a detection rate exceeding 97%. This proactive approach shifts the focus from reactive URL classification to early-stage threat anticipation. However, the method requires extensive historical and network-level data, which may not be readily available to all organizations.

III. Dataset Description

The dataset for this study was created by integrating a variety of open-source cybersecurity repositories to achieve wide coverage of real-life phishing patterns. It drew primarily on the Malicious URL Dataset, with 651,191 URLs in malicious, phishing, and benign categories that introduce substantial lexical and structural variability. A filtered subset of 165,366 phishing and legitimate URLs from the ISCX-URL2016 repository included well-labeled samples collected from real network environments. Table 1 presents the data description which were imported for training the model.

After preprocessing and cleaning, the final working dataset consisted of 95,912 URLs that were evenly divided into phishing and benign classes for binary classification. These samples contained a wide variety of lexical and statistical features of URLs, thus enabling lightweight analysis without relying on web content.

To alleviate class imbalance and hence improve model generalization, the Synthetic Minority Oversampling Technique was applied to the training partition. After the resampling process, the training set contained a total of 76,814 instances, with equal distribution of phishing and benign URLs, ensuring robust learning across both classes.

Table 1: Dataset Description

Dataset	ISCX-URL2016 Dataset	Data Features Dataset	Malicious URL Dataset
URLs	165,366	95,912	651,191

IV. Methodology

A. Data Analysis

An exploratory assessment of the dataset was carried out to check for integrity in the data and anomalies. Descriptive statistics and visual analyses like histograms, heatmaps, and box plots were used to detect missing values, outliers, and attribute correlations. This initial analysis also helped make further decisions about preprocessing and feature selection.

B. Feature Engineering

1. EDA – Exploratory Data Analysis

EDA was done to extract significant lexical and structural features of URLs, like length, frequency of symbols, entropy, count of subdomains, and usage of HTTPS. The correlation checks were performed in order to remove the redundant attributes, while feature distributions were analyzed visually for the separation of classes.

2. SMOTE-Based Oversampling

Class imbalance was handled by using SMOTE for generating synthetic phishing samples, thus getting a balanced training set. This has improved model generalization, thereby enhancing recall by reducing bias towards legitimate URLs.

C. Baseline Model Evaluation (LazyClassifier)

The performances of different machine learning models, including Logistic Regression, SVM, Naïve Bayes, KNN, Decision Trees, Random Forest, Gradient Boosting, and Extra Trees, were compared with LazyClassifier on equal footing. The experimental results showed that tree-based classifiers performed constantly better, demonstrating a great capability in capturing the complex structure of phishing URLs and benefiting the choice of models for ensemble learning.

D. Ensemble Learning

Ensemble methods were then used to improve the robustness and accuracy of these results. We started by stacking high-performing base models: Random Forest, Gradient Boosting, and XGBoost, whose predictions were refined by a Logistic Regression meta-learner. The soft-voting ensemble aggregated probability outputs from the top-ranked models, attaching higher importance to the stronger performers. Figure 1 illustrates the architecture of the proposed model.

E. Final Model: Champion and EnsembleClassifier

Accuracy, precision, recall, F1-score, ROC-AUC, and the components of a confusion matrix were considered for all classifiers. Random Forest together with the soft-voting ensemble reached the best general results. Random Forest was chosen as the Champion Model because of its good balance in the metrics and its consequent behavior on unseen data, being at the core of the final detection system backed by ensemble integration for improved reliability.

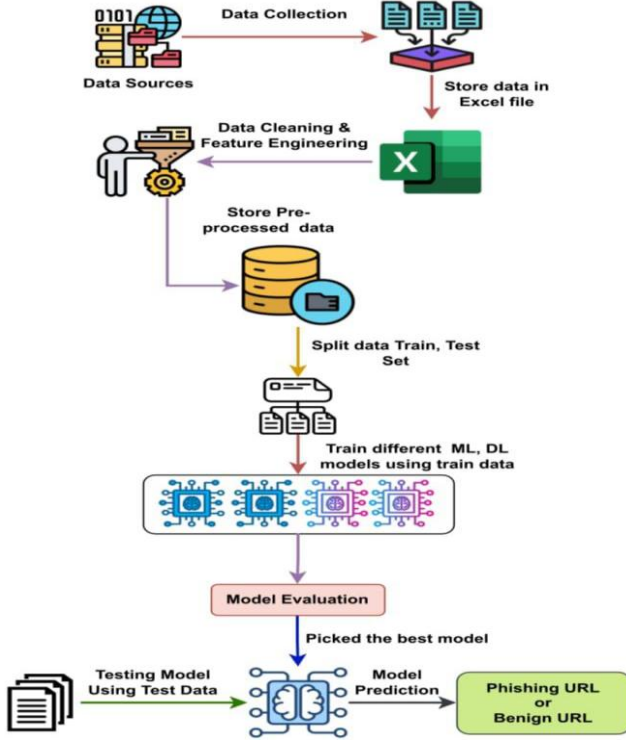


Figure 1: System Architecture

V. Use of Machine Learning Classifiers

The complementary role of machine learning algorithms in detecting phishing URLs is to learn patterns at lexical, structural, and domain levels that differentiate malicious URLs from legitimate ones. Table 2 presents the performance comparison of individual machine learning classifiers based on accuracy, precision, and recall. Logistic Regression is a quick and interpretable linear baseline that offers probabilistic outputs for easy application in initial benchmarking. Naïve Bayes, with a theoretical foundation in Bayes' theorem with independence assumptions, efficiently estimates class probabilities and does comparably well with high-

dimensional lexical features that are commonplace in URL datasets.

Some more sophisticated techniques aim to capture nonlinear relationships and complicated structures of URLs. K-Nearest Neighbors classifies samples by local patterns of similarity within the feature space, while Extra Trees uses randomly highly perturbed decision trees to reduce variance and increase training speed. Gradient Boosting elicits exceptional performance through iteratively correcting mistakes from the previous weak learner, thus becoming very effective for fine-grained pattern refinement. Support Vector Machines find the maximum margin of the decision boundary between classes, allowing strong classification performance based on structured and moderately high-dimensional features.

Deep learning-based approaches go a step further, enhancing detection through automated representation learning. CNNs identify the spatial interactions of tokens in URLs, while LSTMs capture sequential dependencies and transformer-based models leverage self-attention in order to learn long-range and global relationships across an entire URL string.

Among all these methods, Random Forest remains one of the most dependable classifiers for phishing detection. It reduces overfitting, improves generalization of the models, and effectively models the complex nonlinear patterns present in large-scale URL datasets by aggregating multiple decision trees via bootstrap. Its robustness and consistent performance make it a perfect fit for practical phishing detection systems.

Table 2: Individual Classifier Performance

Model	Accuracy	Precision	Recall
Random Forest	0.95	0.96	0.95
Extra Trees	0.95	0.96	0.95
Bagging Classifier	0.95	0.96	0.94
K-Nearest Neighbors	0.92	0.93	0.91
Gradient Boosting	0.90	0.91	0.89
Logistic Regression	0.66	0.61	0.86

VI. Random Forest Algorithm

The Random Forest (RF) algorithm is an ensemble-based machine learning technique that constructs multiple decision trees using bootstrap sampling and random feature selection. By training each tree on different subsets of the data and features, Random Forest effectively reduces overfitting while improving generalization. Figure 2 illustrates the confusion matrix obtained during the training phase of the Random Forest classifier. This makes it particularly suitable for phishing URL detection, where the feature space is large, nonlinear, and often noisy. Its ability to model complex interactions among features provides a strong advantage over single-tree or linear classifiers.

In the phishing detection framework, Random Forest evaluates a wide range of lexical and domain-level attributes, including IP address presence, subdomain structure, domain age, traffic ranking, DNS records, Google index status, and other URL-related clues. Each decision tree analyzes these attributes independently, and their outputs are aggregated through majority voting to determine whether a URL is phishing or legitimate. This ensemble strategy ensures classification stability and improves resilience against adversarial variations commonly found in phishing URLs.

During experimentation, Random Forest emerged as the highest-performing model, achieving an F1-score of 0.96 with strong confusion matrix results—True Positives: 9086 and True Negatives: 9244. Its robustness to noise, high-dimensional feature sets, and hybrid lexical-domain features contributed to consistently superior accuracy and recall compared to other classifiers. Figure 3 illustrates the confusion matrix obtained during the testing phase of the Random Forest classifier. Furthermore, the model's feature-importance capabilities provide valuable interpretability, highlighting which URL attributes carry the most predictive influence. These strengths collectively establish Random Forest as the most dependable choice for real-time phishing URL detection in this study.

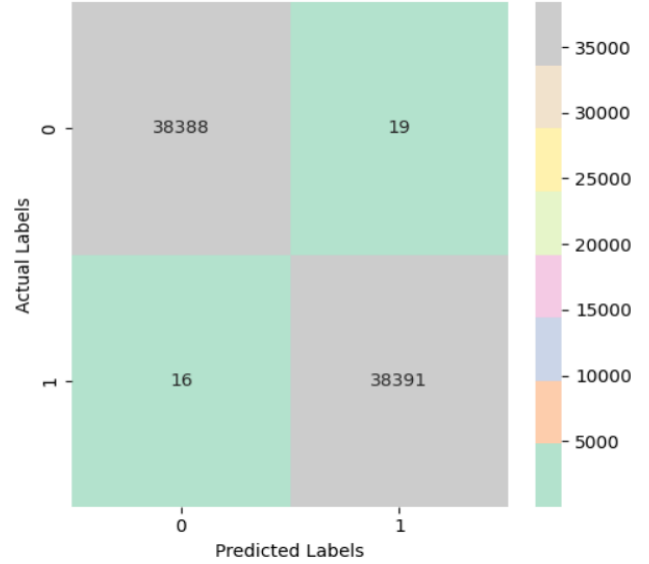


Figure 2: Random Forest Training Confusion Matrix

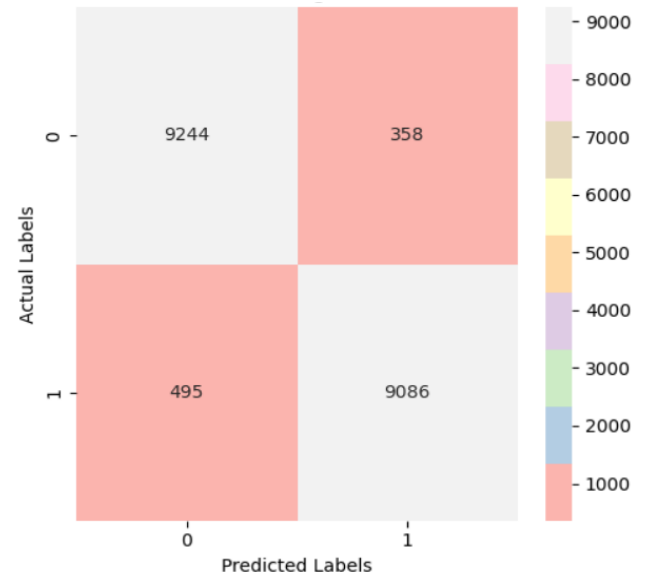


Figure 3: Random Forest Testing Confusion Matrix

VII. Experimental Results and Evaluation

Tree-based models performed distinctly better compared with linear and probabilistic classifiers. Figure 4 and 5 illustrates the performance metrics obtained during the training and testing phase of the Random Forest classifier. Among all compared methods, Random Forest showed the best overall performance. Its balanced precision and recall reflect its ability to identify phishing URLs while keeping the misclassification of legitimate ones as low as possible.

The Soft Voting Ensemble, comprising a number of high-performing classifiers, resulted in steady and consistent predictions. However, with a slightly lower F1-score than the Random Forest, it improved the robustness by combining the strength of various algorithms.

Performance gaps across classifiers reveal that lexical phishing detection benefits substantially from nonlinear models able to capture complex structural patterns. Probabilistic models struggled because of feature interdependencies, while distance-based models exhibited sensitivity to scaling and data distribution. Among the evaluated models, Random Forest achieved the highest F1-score of 0.96, followed closely by Ensemble Classifier, scoring above 0.92. Figure 6 illustrates the result comparison of a Champion Model and Ensemble Classifier.

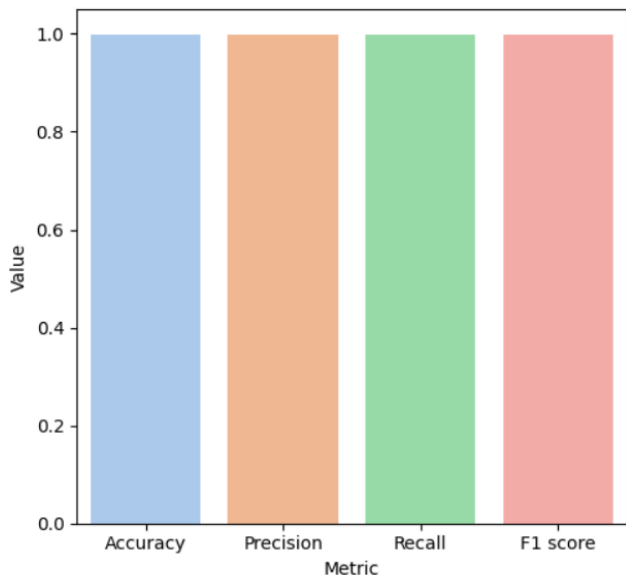


Figure 4: Random Forest Training Performance Metrics

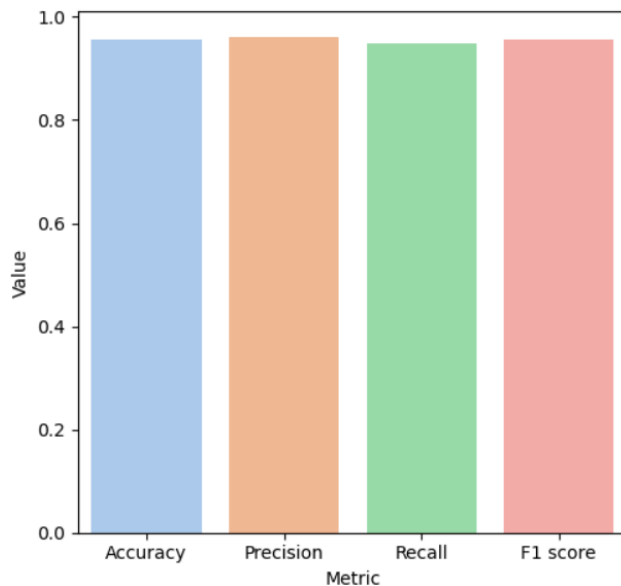


Figure 5: Random Forest Testing Performance Metrics

Comparison of Champion Model and Ensemble Classifier:		
	Model	F1_Score
0	Random Forest	0.96
1	Ensemble Classifier	0.92

Figure 6: Result Comparison of Champion Model and Ensemble Classifier Based on F1 Score

VIII. Discussion

The experimental results demonstrate that tree-based ensemble models, particularly Random Forest and Extra Trees, consistently outperform other classifiers in phishing URL detection. Their ability to handle high-dimensional feature spaces and capture complex decision boundaries contributes to superior precision and recall [1], [4], [10]. The use of SMOTE for data balancing proved effective, eliminating bias toward the majority class and improving generalization [1], [6].

The fusion-based ensemble classifier, while slightly trailing the champion model in F1-score, offers a more balanced bias-variance profile and robustness across TP and TN metrics. This suggests that ensemble fusion is a viable strategy for real-world deployment, where consistent performance across diverse phishing patterns is critical [2], [4], [7].

Despite strong results, the model’s reliance on handcrafted URL features may limit adaptability to evolving attack vectors. Future work could explore dynamic feature extraction using deep learning or transformer-based architectures [3], [8], [9] to enhance resilience against obfuscation and adversarial manipulation.

IX. Research Challenges

A persistent challenge in phishing-detection research is the acute deficit of comprehensive, consistently labelled, and temporally updated datasets [5], [6], [12]. Phishing websites are inherently ephemeral, often disappearing within hours or days, which severely constrains the availability of ground-truth corpora. This volatility results in data sparsity, label drift, and incomplete feature coverage, thereby impeding the development of robust machine-learning models [13], [14]. Moreover, many publicly available repositories (e.g., PhishTank, OpenPhish) exhibit sampling bias, delayed verification cycles, and limited metadata, restricting their utility for high-fidelity inference and generalizable model training [6], [12].

Phishing sites frequently block crawlers, enforce geo-restrictions, employ cloaking techniques, or self-terminate after short lifecycles. The absence of such granular signals forces many studies to rely predominantly on URL-centric lexical features, which limits model expressiveness and increases susceptibility to adversarial perturbations [1], [10]. Legitimate websites vastly outnumber phishing sites, creating severe class imbalance that complicates supervised learning [1], [11]. The imbalance is further exacerbated by the difficulty of sourcing up-to-date phishing samples, leading to skewed distributions and unbalanced training matrices. Standard remedies such as oversampling, SMOTE, or cost-sensitive learning mitigate but do not fully resolve distortions caused by unstable data inflow and constrained phishing-site availability [6], [11].

The phishing landscape evolves at a pace that outstrips the speed at which curated datasets can be updated [5], [9]. Attackers continuously mutate URL structures, obfuscation patterns, keyword embeddings, and hosting infrastructures. Due to the lack of sustained, long-term archival repositories of phishing sites, researchers struggle to conduct longitudinal studies or train models resilient to concept drift and zero-day attack variants [7], [13]. Collecting raw phishing pages introduces legal and ethical complications, especially when dealing with malicious scripts or sensitive personal information embedded by attackers [12], [15]. Many institutional environments impose restrictions on mirroring phishing websites or crawling suspicious links, resulting in institutional

resource shortages and limiting researcher ability to acquire representative malicious samples [13], [14]. This further amplifies the scarcity of high-quality phishing-site resources.

X. Conclusion and Future Work

This work introduced a rigorously engineered phishing-detection architecture underpinned by a multi-modal machine-learning pipeline operating on a large-scale corpus of approximately 737,000 URLs. By synthesizing lexical entropy measures, domain-intelligence metadata, and lightweight client-side artifact signatures, the proposed system demonstrated heightened discriminative capability against highly polymorphic and obfuscated phishing vectors [1], [3], [4].

Empirical benchmarking revealed that ensemble-based classifiers, particularly Random Forest and Gradient Boosting, consistently outperformed traditional heuristic mechanisms by achieving elevated true-positive rates, enhanced recall for malicious entities, and improved robustness under high-variance adversarial conditions [1], [4], [10]. The convergence of feature-rich representations and optimized model tuning established a resilient detection substrate capable of supporting real-time threat interception within SOC infrastructures and security orchestration workflows [5], [9].

Future research trajectories will emphasize the incorporation of deep neural architectures including Transformer-based sequence encoders, CNN-driven visual page-render analysis, and hybrid embedding stacks to capture latent semantic and behavioural cues that static URL features cannot encode [3], [8], [9]. Additional focus will be directed toward adversarial hardening, leveraging techniques such as perturbation-resilient training, anomaly-driven zero-day identification, and generative simulation of evolving threat landscapes [5], [7].

Furthermore, operational scalability can be advanced through cloud-native microservices, containerized low-latency inference pipelines, and continuous learning mechanisms that autonomously recalibrate model parameters to mitigate drift [6], [13]. Collectively, these enhancements aim to evolve the framework into a fully autonomous, adaptive, and intelligence-centric cyber-defence ecosystem, capable of neutralizing next-generation phishing campaigns with precision, agility, and Operational resilience.

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